## Method of expressing concentration of solution

**101.** When $7.1 gm Na_2SO_4$ (molecular mass 142) dissolves in $100 ml H_2O$, the molarity of the solution is
(a) 2.0 *M* (b) 1.0 *M*
(c) 0.5 *M* (d) 0.05 *M*

**102.** Molarity of 4% $NaOH$ solution is
(a) $0.1M$ (b) $0.5M$
(c) $0.01M$ (d) $1.0M$

**103.** When $6gm$ urea dissolve in $180 gm H_2O$. The mole fraction of urea is
(a) $\frac{10}{10.1}$ (b) $\frac{10.1}{10}$
(c) $\frac{10.1}{0.1}$ (d) $\frac{0.1}{10.1}$

**104.** The normality of 10% (weight/volume) acetic acid is
(a) 1 N (b) 10 N
(c) 1.7 N (d) 0.83 N

**105.** Unit of mole fraction is
(a) *Moles/litre*
(b) *Moles/litre*$^2$
(c) Moles–litre
(d) Dimensionless

**106.** Normality of $2M$ sulphuric acid is
(a) $2N$ (b) $4N$
(c) $N/2$ (d) $N/4$

**107.** Molar concentration $(M)$ of any solution =
(a) $\frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$
(b) $\frac{\text{No. of gram equivalent of solute}}{\text{Volume of solution in litre}}$
(c) $\frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$
(d) $\frac{\text{No. of moles of any constituent}}{\text{Total no. of moles of all constituents}}$

**108.** If $5.0gm$ of $BaCl_2$ is present in $10^6 gm$ solution, the concentration is
(a) 1 *ppm* (b) 5 *ppm*
(c) 50 *ppm* (d) 1000 *ppm*

**109.** 1 Molar solution contains
(a) 1000*g* of solute
(b) 1000*g* of solvent
(c) 1 *litre* of solvent
(d) 1 *litre* of solution

**110.** To neutralise completely 20 mL of 0.1 M aqueous solution of phosphorous acid $(H_3PO_3)$, the volume of 0.1 M aqueous *KOH* solution required is
(a) 40 *mL* (b) 20 *mL*
(c) 10 *mL* (d) 60 *mL*

**111.** On dissolving 1 mole of each of the following acids in 1 litre water, the

acid which does not give a solution of strength $1N$ is
(a) $HCl$
(b) Perchloric acid
(c) $HNO_3$
(d) Phosphoric acid

**112.** How many grams of $NaOH$ will be required to neutralize 12.2 grams of benzoic acid
(a) $40gms$ (b) $4gms$
(c) $16gms$ (d) $12.2gms$

**113.** $10ml$ of conc. $H_2SO_4$ (18 molar) is diluted to 1 litre. The approximate strength of dilute acid could be
(a) 0.18 *N* (b) 0.09 *N*
(c) 0.36 *N* (d) 1800 *N*

**114.** The normality of 10 *lit.* volume hydrogen peroxide is
(a) 0.176 (b) 3.52
(c) 1.78 (d) 0.88
(e) 17.8

**115.** Essential quantity of ammonium sulphate taken for preparation of 1 molar solution in 2 litres is
(a) $132gm$ (b) $264gm$
(c) $198gm$ (d) $212gm$

**116.** In a mixture of 1 *gm* $H_2$ and 8 *gm* $O_2$, the mole fraction of hydrogen is
(a) 0.667
(b) 0.5
(c) 0.33
(d) None of these

**117.** A solution of $CaCl_2$ is $0.5mol/litre$, then the moles of chloride ion in $500ml$ will be
(a) 0.25 (b) 0.50
(c) 0.75 (d) 1.00

**118.** What is the molarity of $H_2SO_4$ solution, that has a density 1.84 *gm/cc* at $35^oC$ and contains solute 98% by weight
(a] 4.18 *M* (b) 8.14 *M*
(c) 18.4 *M* (d) 18 *M*

**119.** A certain aqueous solution of $FeCl_3$ (formula mass =162) has a density of $1.1g/ml$ and contains $20.0\%FeCl_3$. Molar concentration of this solution is
(a) 0.028 (b) 0.163
(c) 1.27 (d) 1.47

**120.** If 0.50 mol of $CaCl_2$ is mixed with 0.20 mol of $Na_3PO_4$, the maximum number of moles of $Ca_3(PO_4)_2$ which can be formed, is
(a) 0.70 (b) 0.50
(c) 0.20 (d) 0.10

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